

Graded All-Orders Asymptotics for Successive-Primorial Crossovers in the Odlyzko–Rubinstein–Wolf Exponential-Integral Approximation

Mohamed Hassan Ali Awaad Saraya

Abstract

Let $P = p_k^\#$ and $q = p_{k+1}$, so that $qP = p_{k+1}^\#$ is the next primorial. We study the pairwise crossover between these two candidates in the explicit Odlyzko–Rubinstein–Wolf (ORW) exponential-integral approximation

$$\mathcal{M}(X, D) = \mathfrak{S}(D)I_D(X), \quad I_D(X) = \int_2^X \frac{\exp(-D/\log t)}{(\log t)^2} dt.$$

Writing $L = \log X$, we prove a uniform all-orders endpoint expansion for $I_D(e^L)$ whose coefficients are $j!L_j^{(1)}(D/L)$, with remainder control stable under differentiation in the growing endpoint parameter. We then reorganize the logarithmic expansion into a graded asymptotic scale and rigorously invert the full crossover equation. If

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}$$

is the exact solution of ORW's leading exponential comparison, then for every fixed M

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(P, qP)}{L_0^m} + O_M\left(\frac{(1+qP)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}}\right),$$

where $\deg c_m \leq \lfloor m/2 \rfloor$ and the coefficients are generated by an explicit Lagrange–Bürmann formula. In particular,

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1+qP}{L_0^2}\right).$$

The corrections 1 and $3/L_0$ are universal for general endpoint pairs with $D_1/L_0 \rightarrow 0$. The results are theorems about the stated ORW approximation; no unconditional claim about actual jumping champions is made.

Keywords: jumping champions; prime gaps; primorials; generalized Laguerre polynomials; asymptotic inversion.

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1 Introduction and main results

Let

$$2 = p_1 < p_2 < p_3 < \cdots$$

be the primes, and for an even integer D let

$$N(X, D) = \#\{n : p_{n+1} \leq X, p_{n+1} - p_n = D\}.$$

A value of D maximizing $N(X, D)$ is called a *jumping champion*. Erdős and Straus [1] studied the most frequent differences between consecutive primes, and Odlyzko, Rubinstein, and Wolf (ORW) [2] later proposed a detailed heuristic model and numerical picture in which primorials govern the successive champions. Rigorous conditional results in this direction were developed by Goldston and Ledoan [3, 4, 6] and, for k -tuples, by Feng and Wu [7].

This paper isolates one analytic question inside the explicit ORW model. Put

$$P = p_k^\#, \quad q = p_{k+1}, \quad D_0 = P, \quad D_1 = qP.$$

We determine, to every fixed asymptotic order, the pairwise crossover predicted by

$$\mathcal{M}(X, D) = \mathfrak{S}(D)I_D(X), \quad I_D(X) = \int_2^X \frac{\exp(-D/\log t)}{(\log t)^2} dt.$$

The difficulty is not formal expansion by itself. At the relevant crossover the endpoint parameter grows with the primorial, so a fixed- D expansion does not provide the uniform remainder information needed for rigorous root inversion.

Endpoint expansions for Laplace-type integrals and Watson's lemma are classical; standard references include Olver [10] and Wong [11]. The first result here is tailored to the ORW integral: after the endpoint change of variables, the classical Laguerre generating function packages all coefficients as

$$j!L_j^{(1)}(D/L),$$

and the remainder remains uniform after any fixed number of derivatives in the scaled endpoint parameter. This differentiated control is what permits the growing- D crossover equation to be inverted without treating D as fixed.

The ORW leading exponential comparison between P and qP has the exact logarithmic solution

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}.$$

This scale is already implicit in ORW and is not claimed as new. Our main theorem proves that the root L_{int} of the *full* ORW integral comparison has, for every fixed M , the graded expansion

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M\left(\frac{(1+D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}}\right),$$

with

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

The coefficients are generated by an explicit Lagrange–Bürmann formula. The first terms are

$$c_0 = 1, \quad c_1 = 3, \quad c_2 = 12 - \frac{D_0 + D_1}{2}, \quad c_3 = 63 - 4(D_0 + D_1).$$

Moreover, for an arbitrary endpoint pair $D_0 < D_1$ satisfying $L_0 \rightarrow \infty$ and $D_1/L_0 \rightarrow 0$,

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1+D_1}{L_0^2}\right).$$

Thus the first two corrections are universal; dependence on the endpoint pair enters at the next polynomial coefficient.

There is also a separate algebraic hierarchy already contained in the leading ORW scale:

$$\frac{L_0}{P} = q^2 - \frac{5}{2}q + \frac{17}{12} - \frac{1}{24q} - \frac{49}{720q^2} - \frac{163}{1440q^3} + \cdots.$$

Neither this expansion nor its coefficient $-5/2$ is presented as an independent novelty claim. The new point is the all-orders treatment of the full endpoint integral around this scale and the resulting absolute displacement beginning with $+1$.

1.1 Priority and scope

Precise transition locations are already part of the ORW discussion, and later work of Goldston and Ledoan addresses related small-normalized-gap regimes [4]; they also announced a sequel containing sharper formulas [5]. Generalized Laguerre polynomials likewise have precedent in prime-gap research: Goldston and Yıldırım use them in a different orthogonal-polynomial optimization problem [8]. Lagrange–Bürmann inversion and endpoint asymptotics are, of course, classical tools.

Accordingly, the novelty claim is deliberately narrow: the uniform differentiated Laguerre endpoint expansion in the growing-endpoint ORW regime, the graded all-orders inversion of the full successive-primorial crossover, the associated degree hierarchy and explicit coefficient generator, and the universal displacement $1 + 3/L_0$. To the best of our search of the published and publicly accessible literature, we have not located this combination in the ORW or jumping-champion literature; we do not claim absolute historical priority beyond that statement.

The distinction between the ORW approximation and actual consecutive-prime gaps is essential. The theorems below concern the explicitly defined function $\mathcal{M}(X, D)$. Transferring the same precision to $N(X, D)$ requires additional uniform prime-correlation information and is not proved here. Section 4 briefly records this boundary of the result rather than developing a separate conditional theory.

2 The ORW exponential-integral approximation

For even D , write

$$\mathfrak{S}(D) = 2C_2 \prod_{\substack{p|D \\ p>2}} \frac{p-1}{p-2}.$$

For successive primorials,

$$\frac{\mathfrak{S}(qP)}{\mathfrak{S}(P)} = \frac{q-1}{q-2}. \quad (2.1)$$

Define

$$\mathcal{M}(X, D) = \mathfrak{S}(D)I_D(X), \quad I_D(X) = \int_2^X \frac{e^{-D/\log t}}{(\log t)^2} dt,$$

and set $L = \log X$. The prime number theorem in Chebyshev form gives

$$\log P = \vartheta(p_k) \sim p_k \sim q, \quad P = e^{(1+o(1))q}. \quad (2.2)$$

The transition scale obtained below satisfies

$$L \asymp Pq^2, \quad q \sim \log L \sim \log \log X. \quad (2.3)$$

In particular, for every fixed $A > 0$,

$$L^{-1} = o(q^{-A}). \quad (2.4)$$

The use of an endpoint expansion here belongs to the classical Watson–Laplace framework; see, for example, Olver [10] and Wong [11]. What is needed below, and proved in Theorem 2.1, is a form uniform in the growing endpoint parameter together with differentiated remainder control.

2.1 A generating-function proof and differentiated remainder

The generalized Laguerre structure in the endpoint expansion is not merely an algebraic consequence of repeated differentiation. It is already present in the integral through the classical Laguerre generating function.

Put

$$z = \frac{1}{L}, \quad x = \frac{D}{L}.$$

With $u = L - s$, one obtains

$$I_D(e^L) = \frac{e^L e^{-x}}{L^2} \mathcal{J}(z, x),$$

where

$$\mathcal{J}(z, x) = \int_0^{L-\log 2} e^{-s} (1 - sz)^{-2} \exp\left(-x \left((1 - sz)^{-1} - 1\right)\right) ds.$$

Since

$$(1 - sz)^{-1} - 1 = \frac{sz}{1 - sz},$$

the classical generating function

$$\sum_{j \geq 0} L_j^{(1)}(x) t^j = (1 - t)^{-2} \exp\left(-\frac{xt}{1 - t}\right)$$

gives, formally with $t = sz$,

$$(1 - sz)^{-2} \exp\left(-x \frac{sz}{1 - sz}\right) = \sum_{j \geq 0} L_j^{(1)}(x) (sz)^j.$$

Theorem 2.1 (All-orders endpoint expansion and differentiated remainder). *Fix integers $J \geq 1$ and $r \geq 0$, and fix $0 < \eta < 1$. Uniformly for $0 \leq x \leq \eta$ as $z \rightarrow 0^+$,*

$$\frac{\partial^r}{\partial x^r} \left[\mathcal{J}(z, x) - \sum_{j=0}^{J-1} j! L_j^{(1)}(x) z^j \right] = O_{\eta, J, r}(z^J).$$

Consequently,

$$I_D(e^L) = \frac{e^L e^{-D/L}}{L^2} \left[\sum_{j=0}^{J-1} \frac{j! L_j^{(1)}(D/L)}{L^j} + O_{\eta, J}(L^{-J}) \right],$$

and the normalized remainder may be differentiated with respect to D , with each D -derivative gaining one further factor L^{-1} .

Proof. Split the s -integral at

$$S = z^{-1/2}.$$

On $0 \leq s \leq S$, one has

$$|sz| \leq z^{1/2},$$

so for sufficiently small z the Laguerre generating function is analytic in a fixed disk containing every $t = sz$ under consideration. Taylor's theorem, uniformly for $x \in [0, \eta]$ and after any fixed number r of x -derivatives, gives

$$\frac{\partial^r}{\partial x^r} \left[(1 - sz)^{-2} \exp \left(-x \frac{sz}{1 - sz} \right) - \sum_{j=0}^{J-1} L_j^{(1)}(x) (sz)^j \right] = O_{\eta, J, r}((sz)^J).$$

Multiplying by e^{-s} and integrating yields

$$O_{\eta, J, r} \left(z^J \int_0^\infty e^{-s} s^J ds \right) = O_{\eta, J, r}(z^J).$$

On the complementary interval

$$S \leq s \leq z^{-1} - \log 2,$$

one has

$$1 - sz \geq z \log 2.$$

The exponential factor involving x has absolute value at most one for $x \geq 0$, while an x -derivative of order r introduces at most the factor

$$\left(\frac{sz}{1 - sz} \right)^r = O_r(z^{-r}).$$

Together with $(1 - sz)^{-2} = O(z^{-2})$, the tail is therefore bounded by

$$O_r \left(z^{-r-2} \int_{z^{-1/2}}^\infty e^{-s} ds \right) = O_r \left(z^{-r-2} e^{-z^{-1/2}} \right) = O_{r, A}(z^A)$$

for every fixed $A > 0$. The same estimate applies to each term of the finite Laguerre truncation, so the complementary interval is negligible compared with z^J .

Finally,

$$\int_0^\infty e^{-s} s^j ds = \Gamma(j + 1) = j!,$$

which gives the claimed coefficients.

Since $x = Dz$,

$$\frac{\partial}{\partial D} = z \frac{\partial}{\partial x}.$$

Thus every D -derivative of the normalized remainder gains one additional factor $z = L^{-1}$. $\square$

Remark 2.2. *Theorem 2.1 supplies the uniform differentiated remainder required in the proof of the graded all-orders root theorem. In particular, when the normalized endpoint remainders at D_1 and D_0 are subtracted and divided by $A = D_1 - D_0$, the mean value theorem contributes one D -derivative and hence one additional factor L^{-1} . This prevents the endpoint remainder from contaminating the coefficient scale in the Lagrange inversion.*

Remark 2.3 (Role of the Laguerre representation). *The occurrence of Laguerre polynomials in Theorem 2.1 is a compact organization of an endpoint asymptotic expansion, rather than a claim that Laguerre asymptotics themselves are new. The point needed later is that the generating-function representation gives all coefficients together with differentiated uniform remainders, which remain usable when the endpoint parameter D grows with the successive primorials.*

2.2 Monotonicity, uniqueness, and localization of the crossover

Let

$$w(t) = \frac{e^{-P/\log t}}{(\log t)^2}, \quad r(t) = e^{-(q-1)P/\log t}.$$

Then

$$\frac{I_{qP}(X)}{I_P(X)} = \frac{\int_2^X w(t)r(t) dt}{\int_2^X w(t) dt}.$$

The function $r(t)$ is strictly increasing. Hence the ratio above is a weighted average strictly below $r(X)$, and differentiation gives the following.

Lemma 2.4 (Monotonicity). *The map*

$$X \mapsto \frac{I_{qP}(X)}{I_P(X)}$$

is strictly increasing. Therefore the model crossover, if it exists, is unique.

Put

$$A = (q-1)P, \quad s = \log \frac{q-1}{q-2}, \quad L_0 = \frac{A}{s}, \quad S = (q+1)P. \quad (2.5)$$

From the logarithmic endpoint coefficients derived below,

$$\begin{aligned} F(L) &:= \log \frac{\mathcal{M}(e^L, qP)}{\mathcal{M}(e^L, P)} \\ &= s - \frac{A}{L} - \frac{A}{L^2} - \frac{4A}{L^3} + \frac{A(S/2 - 22)}{L^4} + \frac{A(6S - 148)}{L^5} + O\left(\frac{A(1 + S^2)}{L^6}\right). \end{aligned} \quad (2.6)$$

Since $s = A/L_0$, this gives

$$F(L_0) = -\frac{A}{L_0^2}(1 + o(1)) < 0$$

and

$$F(L_0 + 2) = \frac{A}{L_0^2}(1 + o(1)) > 0.$$

Consequently, for sufficiently large q ,

$$L_0 < L_{\text{int}} < L_0 + 2. \quad (2.7)$$

This a priori localization is used before the graded root inversion and prevents circularity.

3 All-orders inversion of the successive-primorial crossover

3.1 Graded endpoint expansion and rigorous root inversion

Put

$$z = \frac{1}{L}, \quad D_0 = P, \quad D_1 = qP, \quad A = D_1 - D_0 = (q-1)P,$$

and

$$t = \frac{1}{L_0} = \frac{1}{A} \log \frac{q-1}{q-2}.$$

For the root analysis, reparameterize the normalized endpoint integral by

$$\mathcal{J}_D(z) = \frac{I_D(e^{1/z})}{e^{1/z} e^{-Dz} z^2}.$$

Equivalently,

$$\mathcal{J}_D(z) = \int_0^{z^{-1} - \log 2} e^{-s} (1 - sz)^{-2} \exp\left(-Dz \frac{sz}{1 - sz}\right) ds.$$

By Theorem 2.1, for every fixed J ,

$$\mathcal{J}_D(z) = \sum_{j=0}^{J-1} j! L_j^{(1)}(Dz) z^j + O_J(z^J)$$

uniformly whenever $0 \leq Dz \leq \eta < 1$. The same remainder may be differentiated with respect to D , with one additional factor z .

Define the formal series

$$\mathcal{B}_D(z) = \sum_{j \geq 0} j! L_j^{(1)}(Dz) z^j = 1 + \sum_{n \geq 1} b_n(D) z^n$$

and

$$\mathcal{H}_D(z) = \log \mathcal{B}_D(z) = \sum_{n \geq 1} h_n(D) z^n.$$

Lemma 3.1 (Explicit endpoint coefficients). *For every $n \geq 1$,*

$$b_n(D) = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m \frac{(n-m)!}{m!} \binom{n-m+1}{m+1} D^m.$$

Hence

$$\deg_D b_n \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. Use

$$L_j^{(1)}(x) = \sum_{m=0}^j (-1)^m \binom{j+1}{m+1} \frac{x^m}{m!}.$$

The contribution indexed by (j, m) is proportional to $D^m z^{j+m}$. Setting $j = n - m$ gives the formula, and $m \leq j$ is equivalent to $m \leq n/2$. $\square$

Lemma 3.2 (Logarithmic coefficient recurrence). *Let $b_0 = 1$. Then*

$$h_n = b_n - \frac{1}{n} \sum_{k=1}^{n-1} k h_k b_{n-k}, \quad n \geq 1,$$

and

$$\deg_D h_n \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Proof. From

$$\mathcal{B}'_D(z) = \mathcal{H}'_D(z)\mathcal{B}_D(z)$$

compare the coefficient of z^{n-1} .

For the degree bound, every contribution to h_n arising from $\log(1+U)$ is a product

$$b_{n_1} \cdots b_{n_r}, \quad n_1 + \cdots + n_r = n.$$

Its degree is at most

$$\sum_i \left\lfloor \frac{n_i}{2} \right\rfloor \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

□

The first coefficients are

$$h_1(D) = 2,$$

$$h_2(D) = 4 - D,$$

$$h_3(D) = \frac{44}{3} - 4D,$$

$$h_4(D) = 74 - 22D + \frac{1}{2}D^2,$$

$$h_5(D) = \frac{2312}{5} - 148D + 6D^2,$$

and

$$h_6(D) = \frac{10192}{3} - 1156D + 64D^2 - \frac{1}{3}D^3.$$

Lemma 3.3 (Graded logarithmic remainder). *Fix $N \geq 1$ and $0 < \eta < 1$. Let*

$$r_N = \left\lfloor \frac{N+1}{2} \right\rfloor.$$

Uniformly in the regime

$$0 < z \rightarrow 0, \quad 0 \leq Dz \leq \eta,$$

one has

$$\log \mathcal{J}_D(z) = \sum_{n=1}^N h_n(D)z^n + O_{N,\eta}\left((1+D)^{r_N}z^{N+1}\right).$$

Moreover,

$$\frac{\partial}{\partial D} \left[\log \mathcal{J}_D(z) - \sum_{n=1}^N h_n(D)z^n \right] = O_{N,\eta}\left((1+D)^{r_N-1}z^{N+1}\right).$$

Proof. Apply Theorem 2.1 with $J = N+1$. Writing

$$P_N(D, z) = \sum_{j=0}^N j! L_j^{(1)}(Dz) z^j,$$

we have

$$\mathcal{J}_D(z) = P_N(D, z) + R_N(D, z),$$

where, uniformly for $Dz \leq \eta$,

$$R_N(D, z) = O_{N,\eta}(z^{N+1}), \quad \partial_D R_N(D, z) = O_{N,\eta}(z^{N+2}). \quad (3.1)$$

The second estimate is not obtained by formally differentiating the first: it is the $r = 1$ case of Theorem 2.1, together with $\partial_D = z\partial_x$.

Expanding the finite Laguerre polynomials gives

$$P_N(D, z) = 1 + \sum_{n=1}^{2N} \tilde{b}_{n,N}(D) z^n,$$

where $\tilde{b}_{n,N} = b_n$ for $n \leq N$ and

$$\deg_D \tilde{b}_{n,N} \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Set

$$B_N(D, z) = 1 + \sum_{n=1}^N b_n(D) z^n.$$

For $n \geq N+1$, the graded monomial sizes

$$(1+D)^{\lfloor n/2 \rfloor} z^n$$

have successive ratios equal to either z or $(1+D)z$. Since $(1+D)z \leq \eta + z < 1$ for all sufficiently small z , the omitted finite polynomial tail is dominated by its first graded scale. Hence

$$\mathcal{J}_D(z) - B_N(D, z) = O_{N,\eta}\left((1+D)^{r_N} z^{N+1}\right). \quad (3.2)$$

Differentiating the *finite* omitted polynomial tail lowers every positive degree in D by one; combining this with (3.1) gives

$$\partial_D(\mathcal{J}_D(z) - B_N(D, z)) = O_{N,\eta}\left((1+D)^{r_N-1} z^{N+1}\right). \quad (3.3)$$

(When an omitted coefficient is independent of D , its derivative is simply zero.)

Both $\mathcal{J}_D(z)$ and $B_N(D, z)$ equal $1 + O_{N,\eta}(z)$ uniformly. Therefore

$$\log \mathcal{J}_D(z) - \log B_N(D, z) = O_{N,\eta}\left((1+D)^{r_N} z^{N+1}\right). \quad (3.4)$$

For the derivative, use the exact identity

$$\partial_D(\log \mathcal{J}_D - \log B_N) = \frac{\partial_D(\mathcal{J}_D - B_N)}{\mathcal{J}_D} - \frac{(\mathcal{J}_D - B_N) \partial_D B_N}{\mathcal{J}_D B_N}. \quad (3.5)$$

The first term has the size in (3.3). In the second term, $\partial_D B_N = O_{N,\eta}(z^2)$: indeed differentiation lowers the D -degree of each b_n , and every resulting monomial can be written as $(Dz)^m z^{n-m}$ with $n-m \geq 2$. Thus the second term is smaller than the bound in (3.3).

Finally, because B_N is a finite polynomial with the graded degree bounds already proved, the Taylor expansion of $\log B_N$ through order N is

$$\sum_{n=1}^N h_n(D) z^n,$$

and its omitted terms, as well as their D -derivatives, satisfy the same two graded bounds by the identical successive-ratio argument. Combining this observation with (3.4)–(3.5) proves both assertions. $\square$

For $n \geq 2$, define

$$a_n(D_0, D_1) = \frac{h_n(D_1) - h_n(D_0)}{D_1 - D_0}.$$

Then a_n is symmetric and

$$\deg a_n \leq \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

Define

$$\Phi(z) = z - \sum_{n \geq 2} a_n z^n.$$

For the successive primorial pair,

$$\begin{aligned} \Phi(z) = & z + z^2 + 4z^3 + \left(22 - \frac{S}{2}\right)z^4 + (148 - 6S)z^5 \\ & + \left(1156 - 64S + \frac{Q}{3}\right)z^6 + \cdots, \end{aligned}$$

where

$$S = D_0 + D_1$$

and

$$Q = D_0^2 + D_0 D_1 + D_1^2.$$

Since $\Phi(z) = z + O(z^2)$, it possesses a unique formal inverse (see, e.g., [9])

$$\Psi(t) = \Phi^{\langle -1 \rangle}(t) = \sum_{n \geq 1} \psi_n t^n.$$

Lemma 3.4 (Inverse-coefficient degree bound). *For $n \geq 2$,*

$$\deg \psi_n \leq \left\lfloor \frac{n-2}{2} \right\rfloor.$$

Proof. Put

$$G(u) = \frac{\Phi(u)}{u} = 1 + \sum_{r \geq 1} g_r u^r.$$

Then

$$g_r = -a_{r+1}, \quad \deg g_r \leq \left\lfloor \frac{r-1}{2} \right\rfloor.$$

Lagrange inversion gives

$$\psi_n = \frac{1}{n} [u^{n-1}] G(u)^{-n}.$$

For $n \geq 2$, every nonconstant monomial contributing to the coefficient of u^{n-1} contains factors $g_{r_1}, \dots, g_{r_\ell}$ with

$$r_1 + \cdots + r_\ell = n - 1.$$

Hence its degree is at most

$$\sum_i \left\lfloor \frac{r_i - 1}{2} \right\rfloor \leq \left\lfloor \frac{n-2}{2} \right\rfloor.$$

□

Lemma 3.5 (Graded truncation of the formal inverse). *Let*

$$\Phi_K(z) = z - \sum_{n=2}^K a_n z^n, \quad Z_K(t) = \sum_{n=1}^K \psi_n t^n,$$

where the ψ_n are the coefficients of the formal inverse of Φ . Then, for every fixed $K \geq 2$,

$$t - \Phi_K(Z_K(t)) = O_K\left((1 + D_1)^{\lfloor (K-1)/2 \rfloor} t^{K+1}\right)$$

in the successive-primorial regime $D_1 t \rightarrow 0$.

Proof. Formal inversion gives cancellation of every coefficient of t^r for $2 \leq r \leq K$. Assign to a coefficient multiplying t^r the polynomial degree bound

$$\deg_D \leq \left\lfloor \frac{r-2}{2} \right\rfloor.$$

To see explicitly that the bound is preserved under composition, consider a monomial from $a_j Z_K(t)^j$, with indices $n_1 + \dots + n_j = r$. We have

$$2 \deg a_j \leq j - 2,$$

and, if s of the indices n_i are at least 2,

$$2 \sum_i \deg \psi_{n_i} \leq \sum_{n_i \geq 2} (n_i - 2).$$

Therefore

$$2 \deg \left(a_j \prod_i \psi_{n_i} \right) \leq j - 2 + \sum_{n_i \geq 2} (n_i - 2) \leq r - 2.$$

Thus every coefficient multiplying t^r in the finite composition has degree at most $\lfloor (r-2)/2 \rfloor$. Hence the coefficient of t^r , $r \geq K+1$, in the fixed truncation has size

$$O_K\left((1 + D_1)^{\lfloor (r-2)/2 \rfloor}\right).$$

For the corresponding graded terms

$$U_r = (1 + D_1)^{\lfloor (r-2)/2 \rfloor} t^r$$

the successive ratios are either t or $(1 + D_1)t$. Both tend to zero in the successive-primorial regime. Since only finitely many terms arise from the fixed polynomial truncation, they are dominated by the first uncanceled scale $r = K+1$, which proves the claim. $\square$

Definition 3.6 (Graded root scale). *For $m \geq 0$, define*

$$E_m(q) = \frac{(1 + D_1)^{\lfloor m/2 \rfloor}}{L_0^m}.$$

We say that a family $R(q)$ has a graded generalized asymptotic expansion

$$R(q) = \sum_{m \geq 0}^{\text{gr}} R_m(q)$$

with respect to $\{E_m\}$ if, for every fixed M ,

$$R(q) - \sum_{m=0}^M R_m(q) = O_M(E_{M+1}(q)), \quad R_m(q) = O_m(E_m(q)).$$

We use this terminology in the explicit sense defined here: the successive envelopes form an asymptotic scale, and each fixed truncation has a remainder bounded by the next envelope. This is modeled on the standard use of asymptotic scales; see, for example, Olver [10].

Lemma 3.7 (The graded envelopes form an asymptotic scale). *In the successive-primorial regime,*

$$E_{m+1} = o(E_m) \quad (m \geq 0 \text{ fixed}).$$

More precisely,

$$\frac{E_{2r+1}}{E_{2r}} = \frac{1}{L_0},$$

whereas

$$\frac{E_{2r+2}}{E_{2r+1}} = \frac{1 + D_1}{L_0} \sim \frac{1}{q}.$$

Proof. The first identity is immediate. For the second, use

$$D_1 = qP, \quad L_0 \sim Pq^2.$$

Thus both ratios tend to zero. □

It is useful to make the two scales dimensionless. Put

$$\alpha = \frac{D_1}{L_0} \asymp \frac{1}{q}, \quad \beta = \frac{1}{L_0}.$$

A monomial D_1^d/L_0^m can then be written as

$$\alpha^d \beta^{m-d}.$$

The degree restriction $d \leq \lfloor m/2 \rfloor$ confines the root corrections to the cone $m - d \geq d$. The dominant envelopes are therefore interlaced as

$$1, \quad \beta, \quad \alpha\beta, \quad \alpha\beta^2, \quad \alpha^2\beta^2, \quad \alpha^2\beta^3, \quad \dots$$

This is the precise sense in which the root expansion is a two-scale graded hierarchy.

Theorem 3.8 (All-orders graded successive-primorial root expansion). *For every fixed integer $M \geq 0$, there exist symmetric polynomials $c_m(D_0, D_1)$, $0 \leq m \leq M$, such that*

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M\left(\frac{(1 + D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}}\right).$$

Moreover,

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor,$$

and therefore

$$\frac{c_m(D_0, D_1)}{L_0^m} = O_m(E_m(q)).$$

Consequently the displayed formula is a graded generalized asymptotic expansion in the sense of Definition 3.6.

If

$$G(u) = \frac{\Phi(u)}{u},$$

then the coefficients are generated by

$$c_m = \frac{1}{m+1} [u^m] G'(u) \left(\frac{u}{\Phi(u)} \right)^{m+1}.$$

At the formal level, the coefficient generator is equivalently encoded by

$$z_{\text{formal}} = \Psi(t), \quad t = \frac{1}{L_0},$$

followed by formal reciprocal expansion. No convergence of Ψ at $t = 1/L_0$ is asserted or required.

Proof. Let

$$F(L) = \log \frac{\mathfrak{S}(D_1) I_{D_1}(e^L)}{\mathfrak{S}(D_0) I_{D_0}(e^L)}.$$

For successive primorials,

$$\frac{1}{A} \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)} = t.$$

Apply Lemma 3.3 and the mean value theorem in the parameter D . For every fixed $N \geq 2$,

$$\frac{F(1/z)}{A} = t - z + \sum_{n=2}^N a_n z^n + O_N \left((1 + D_1)^{\lfloor (N+1)/2 \rfloor - 1} z^{N+1} \right)$$

uniformly in the root neighborhood. Here

$$D_1 z \asymp q^{-1} \rightarrow 0.$$

Set $N = M + 2$, and let

$$z_M(t) = \sum_{n=1}^{M+2} \psi_n t^n.$$

Apply Lemma 3.5 with $K = M + 2$. Since $D_1 t \asymp q^{-1} \rightarrow 0$, it gives

$$t - \Phi_{M+2}(z_M) = O_M \left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3} \right).$$

The truncated analytic crossover polynomial is exactly $t - \Phi_{M+2}(z)$. Therefore the inverse-truncation estimate may be combined directly with the analytic remainder in the preceding expansion, giving

$$\frac{F(1/z_M)}{A} = O_M \left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3} \right).$$

The previously established localization

$$L_0 < L_{\text{int}} < L_0 + 2$$

implies

$$z_{\text{int}} = t + O(t^2).$$

To control the derivative without differentiating an asymptotic remainder in z , use the exact endpoint identity

$$\frac{d}{dL} I_D(e^L) = \frac{e^{L-D/L}}{L^2}, \quad \frac{d}{dL} \log I_D(e^L) = \frac{1}{\mathcal{J}_D(1/L)}.$$

Therefore

$$\frac{d}{dz} \left(\frac{F(1/z)}{A} \right) = -\frac{z^{-2}}{A} \left(\frac{1}{\mathcal{J}_{D_1}(z)} - \frac{1}{\mathcal{J}_{D_0}(z)} \right).$$

Theorem 2.1 with $J = 2$, together with its D -differentiated remainder, gives uniformly in the root neighborhood

$$\frac{\mathcal{J}_{D_1}(z) - \mathcal{J}_{D_0}(z)}{A} = -z^2 + O(z^3), \quad \mathcal{J}_{D_i}(z) = 1 + O(z).$$

Consequently

$$\frac{d}{dz} \left(\frac{F(1/z)}{A} \right) = -1 + O(z) = -1 + o(1)$$

uniformly in this neighborhood. Hence the derivative is bounded away from zero for sufficiently large q . By the mean value theorem,

$$z_{\text{int}} - z_M = O_M\left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+3}\right).$$

Since both z_{int} and z_M are asymptotic to t , taking reciprocals yields

$$L_{\text{int}} = \frac{1}{z_M} + O_M\left((1 + D_1)^{\lfloor (M+1)/2 \rfloor} t^{M+1}\right).$$

To identify the coefficients, use

$$t = \Phi(\Psi(t))$$

and therefore

$$\frac{t}{\Psi(t)} = G(\Psi(t)).$$

At the level of formal coefficient extraction,

$$\frac{1}{\Psi(t)} = \frac{1}{t} G(\Psi(t))$$

as an identity of formal Laurent series. Since the analytic root has already been matched to every fixed truncation above, this identity is used only to identify its coefficients. Lagrange–Bürmann gives

$$[t^n] G(\Psi(t)) = \frac{1}{n} [u^{n-1}] G'(u) \left(\frac{u}{\Phi(u)} \right)^n,$$

which yields the displayed formula for c_m .

Finally, write

$$G(u) = 1 + \sum_{j \geq 1} g_j u^j, \quad \deg g_j \leq \left\lfloor \frac{j-1}{2} \right\rfloor \leq \left\lfloor \frac{j}{2} \right\rfloor.$$

Hence the coefficient of u^r in $G'(u)$ has degree at most $\lfloor r/2 \rfloor$. Every monomial contributing to the coefficient of u^s in the fixed power $G(u)^{-(m+1)}$ is a product $g_{j_1} \cdots g_{j_\ell}$ with $j_1 + \cdots + j_\ell = s$, and therefore has degree at most

$$\sum_i \left\lfloor \frac{j_i}{2} \right\rfloor \leq \left\lfloor \frac{s}{2} \right\rfloor.$$

For $r + s = m$, the product appearing in $[u^m]$ consequently has degree at most

$$\left\lfloor \frac{r}{2} \right\rfloor + \left\lfloor \frac{s}{2} \right\rfloor \leq \left\lfloor \frac{m}{2} \right\rfloor,$$

which proves

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

□

Proposition 3.9 (General pair extension). *Let $D_0 = D_0(\nu) < D_1 = D_1(\nu)$ be a family of even gaps such that*

$$\mathfrak{S}(D_1) > \mathfrak{S}(D_0).$$

Put

$$A = D_1 - D_0, \quad s = \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)}, \quad L_0 = \frac{A}{s}.$$

Assume

$$L_0 \rightarrow \infty, \quad \frac{D_1}{L_0} \rightarrow 0.$$

Then, for sufficiently large ν , the model equation

$$\mathfrak{S}(D_1)I_{D_1}(e^L) = \mathfrak{S}(D_0)I_{D_0}(e^L)$$

has a unique root L_{int} in

$$L_0 < L_{\text{int}} < L_0 + 2,$$

and, for every fixed $M \geq 0$,

$$L_{\text{int}} = L_0 + \sum_{m=0}^M \frac{c_m(D_0, D_1)}{L_0^m} + O_M \left(\frac{(1 + D_1)^{\lfloor (M+1)/2 \rfloor}}{L_0^{M+1}} \right),$$

with the same symmetric coefficient polynomials as in Theorem 3.8. Moreover the envelopes

$$E_m^{\text{gen}} = \frac{(1 + D_1)^{\lfloor m/2 \rfloor}}{L_0^m}$$

form an asymptotic scale under the present assumptions.

Thus the formal coefficient system is a property of the endpoint integral itself; successive primorials enter by supplying the particular singular-series ratio and the more specific relation $D_1/L_0 \asymp q^{-1}$.

Proof. The monotonicity argument of Section 2 uses only $D_1 > D_0$. From the logarithmic endpoint expansion and the mean value theorem in D , uniformly for $L \asymp L_0$,

$$F(L) = s - \frac{A}{L} - \frac{A}{L^2} + O \left(\frac{A}{L^3} + \frac{AD_1}{L^4} \right).$$

Since $s = A/L_0$, at the left endpoint

$$F(L_0) = -\frac{A}{L_0^2} + o \left(\frac{A}{L_0^2} \right) < 0.$$

At the right endpoint,

$$s - \frac{A}{L_0 + 2} = \frac{2A}{L_0(L_0 + 2)},$$

and therefore

$$F(L_0 + 2) = \frac{A}{L_0^2}(1 + o(1)) > 0.$$

Indeed, the remaining terms are $o(A/L_0^2)$ because $L_0 \rightarrow \infty$ and $D_1/L_0 \rightarrow 0$. Existence follows by continuity and uniqueness from monotonicity, giving the asserted localization. In the proof of Theorem 3.8, replace the primordial identity

$$\frac{1}{A} \log \frac{\mathfrak{S}(D_1)}{\mathfrak{S}(D_0)} = t$$

by the definition $t = s/A = 1/L_0$. Every subsequent estimate uses only $t \rightarrow 0$ and $D_1 t \rightarrow 0$, which are precisely the present assumptions. Finally,

$$\frac{E_{2r+1}^{\text{gen}}}{E_{2r}^{\text{gen}}} = L_0^{-1} \rightarrow 0, \quad \frac{E_{2r+2}^{\text{gen}}}{E_{2r+1}^{\text{gen}}} = \frac{1 + D_1}{L_0} \rightarrow 0,$$

so the displayed envelopes indeed form an asymptotic scale. $\square$

Corollary 3.10 (Universal first endpoint corrections). *Under the hypotheses of Proposition 3.9,*

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1 + D_1}{L_0^2}\right).$$

The constants 1 and 3 are independent of the arithmetic structure of D_0, D_1 .

Proof. The logarithmic endpoint coefficients satisfy

$$h_2(D) = 4 - D, \quad h_3(D) = \frac{44}{3} - 4D.$$

Hence their divided differences are universally

$$a_2 = -1, \quad a_3 = -4.$$

Therefore

$$\Phi(z) = z + z^2 + 4z^3 + O((1 + D_1)z^4),$$

and Lagrange–Bürmann inversion gives

$$c_0 = 1, \quad c_1 = 3.$$

The $M = 1$ remainder in the graded theorem is

$$O\left(\frac{1 + D_1}{L_0^2}\right).$$

$\square$

Corollary 3.11 (First coefficients). *With*

$$S = D_0 + D_1, \quad Q = D_0^2 + D_0 D_1 + D_1^2,$$

the first coefficients are

$$\begin{aligned} c_0 &= 1, & c_1 &= 3, \\ c_2 &= 12 - \frac{S}{2}, \\ c_3 &= 63 - 4S, \\ c_4 &= 420 - \frac{63}{2}S + \frac{Q}{3}, \\ c_5 &= 3390 - 280S - \frac{3}{4}S^2 + 6Q, \end{aligned}$$

and

$$c_6 = \frac{1}{4} \left(S^3 - 2SQ + 336Q - 56S^2 - 11300S + 127008 \right).$$

Remark 3.12 (Why “graded all-orders” is the precise terminology). *The coefficients $c_m(D_0, D_1)$ vary with the asymptotic parameter, so the theorem is not being presented as a one-parameter Poincaré series with fixed numerical coefficients. Instead, the explicit envelopes E_m of Definition 3.6 form an asymptotic scale by Lemma 3.7, and the remainder after every fixed truncation is bounded by the next envelope. This is a generalized asymptotic expansion to all fixed orders. No convergence of the infinite formal inverse is asserted.*

3.2 Comparison with ORW’s leading scale and the two-scale hierarchy

The exact identity

$$\log \frac{q-1}{q-2} = \sum_{n \geq 1} \frac{2^n - 1}{nq^n}$$

gives, by ordinary series inversion,

$$\frac{L_0}{P} = q^2 - \frac{5}{2}q + \frac{17}{12} - \frac{1}{24q} - \frac{49}{720q^2} - \frac{163}{1440q^3} + O(q^{-4}). \quad (3.6)$$

This is an expansion of ORW’s leading exponential comparison, not a new transition mechanism. Indeed, in the present actual-gap notation the leading ORW equality is

$$\frac{q-1}{q-2} \exp\left(-\frac{D_1}{L}\right) = \exp\left(-\frac{D_0}{L}\right),$$

whose exact solution is precisely

$$L = L_0 = \frac{P(q-1)}{\log((q-1)/(q-2))}.$$

Replacing the logarithm by $1/(q-2)$ gives ORW’s rough polynomial scale $P(q-1)(q-2)$. Thus neither L_0 nor the coefficient $-5/2$ in (3.6) is claimed as an independent novelty.

The new quantity is the displacement of the *full* integral root from this leading scale. Theorem 3.8 gives

$$L_{\text{int}} - L_0 = 1 + \frac{3}{L_0} + \frac{12 - S/2}{L_0^2} + \cdots, \quad S = D_0 + D_1. \quad (3.7)$$

These two expansions live on separated scales. If

$$E(q) = \frac{q-1}{\log((q-1)/(q-2))},$$

then $L_0 = PE(q)$, while

$$\frac{L_{\text{int}}}{P} = E(q) + \frac{1}{P} + O\left(\frac{1}{PL_0}\right).$$

Since $P = \exp((1 + o(1))q)$, the absolute +1 endpoint displacement is beyond every fixed algebraic order in q^{-1} after normalization by P . This separation is the reason the algebraic refinement of L_0/P and the endpoint-integral correction should not be conflated.

3.3 Numerical consistency check

For numerical evaluation it is convenient to use

$$\frac{I_D(e^L)}{e^L} = \int_0^{L-\log 2} e^{-s} \frac{e^{-D/(L-s)}}{(L-s)^2} ds.$$

The table compares the numerically computed displacement $\Delta = L_{\text{int}} - L_0$ with successive truncations of the root expansion. The numerical calculation is not used in any proof.

q	Numerical Δ	$1 + 3/L_0$	through L_0^{-2}	through L_0^{-4}
5	1.0349513526	1.0359602591	1.0350981656	1.0349516001
7	1.0029269673	1.0030386926	1.0029278888	1.0029269671
11	1.0001473723	1.0001505150	1.0001473736	1.0001473723
13	1.0000092576	1.0000094168	1.0000092576	1.0000092576
17	1.0000003981	1.0000004030	1.0000003981	1.0000003981

Reproducibility. The numerical values in the table were computed independently of the asymptotic coefficient algebra. We evaluated the normalized endpoint integral after the change of variables

$$\frac{I_D(e^L)}{e^L} = \int_0^{L-\log 2} e^{-s} \frac{e^{-D/(L-s)}}{(L-s)^2} ds$$

using 80-decimal-digit arithmetic and adaptive quadrature. We solved $F(L) = 0$ with the `findroot` routine in `mpmath`, using initial values $L_0 + 0.5$ and $L_0 + 1.5$. Both initial values lie inside the already proved interval $(L_0, L_0 + 2)$. The reported digits are stable under increasing the working precision and quadrature cutoff. A short ancillary script, `numerical_check.py`, reproduces the table and is supplied with the source files. These computations are consistency checks only and are not used in any proof.

4 Scope beyond the ORW model

The function $\mathcal{M}(X, D)$ was introduced to model counts of consecutive-prime gaps, but the all-orders theorem above is an analytic theorem about that explicit approximation. Passing from $\mathcal{M}(X, D)$ to the actual count $N(X, D)$ is a separate arithmetic problem. In particular, the endpoint analysis does not by itself control the exclusion of intermediate primes that turns a prime pair of gap D into a *consecutive* prime pair.

Earlier work of ORW, Goldston–Ledoan, and Feng–Wu describes this arithmetic side through Hardy–Littlewood-type hypotheses and singular-series averages [2, 3, 4, 7]. Those results motivate the model but do not presently supply, in the growing successive-primorial regime considered here, an unconditional error term at the precision of the absolute endpoint displacement proved in Section 3. We therefore keep the two statements separate: Sections 2–3 rigorously determine the crossover of the explicit ORW integral model, while no theorem here asserts the corresponding crossover for actual jumping champions.

This separation is intentional. The analytic contribution can be stated and proved without adding a conditional prime-correlation theory whose hypotheses would carry essentially all of the unresolved arithmetic content. A transfer to actual consecutive gaps would be a distinct result requiring new uniform correlation input.

5 Conclusion

The full ORW endpoint integral has a considerably more rigid local structure than its leading exponential comparison alone reveals. Its endpoint coefficients are organized by generalized Laguerre polynomials with uniform differentiated remainder control. After taking logarithms, the growing endpoint parameter induces a graded coefficient hierarchy; this makes Lagrange–Bürmann inversion rigorous at every fixed order and yields explicit polynomial root coefficients with

$$\deg c_m \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

For successive primorials, the reference point

$$L_0 = \frac{(q-1)P}{\log((q-1)/(q-2))}$$

is the exact ORW leading exponential scale, while the full integral root satisfies

$$L_{\text{int}} = L_0 + 1 + \frac{3}{L_0} + O\left(\frac{1+qP}{L_0^2}\right).$$

The first two corrections are universal for general endpoint pairs with $D_1/L_0 \rightarrow 0$, and the complete fixed-order expansion is generated explicitly. These corrections form an absolute hierarchy that is asymptotically separated from the algebraic q^{-1} expansion of L_0/P .

The result is deliberately confined to the ORW model. Its value is a rigorous all-orders description of that model crossover, together with a uniform analytic mechanism that remains valid when the endpoint grows. Establishing an equally precise statement for actual consecutive-prime gaps requires additional arithmetic input and is not claimed here.

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

No research data were generated or analyzed for this theoretical study. The ancillary script `numerical_check.py`, supplied with the submission, reproduces the numerical consistency table; none of the proofs depends on the numerical computation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to support literature exploration, mathematical cross-checking and proof auditing, ancillary numerical-code preparation and checking, and manuscript organization and language refinement. The author reviewed and edited the resulting material, verified the mathematical claims and cited sources used in the final manuscript, and takes full responsibility for the content of the publication.

References

- [1] P. Erdős and E. G. Straus, *Remarks on the differences between consecutive primes*, Elem. Math. **35** (1980), no. 5, 115–118.
- [2] A. M. Odlyzko, M. Rubinstein, and M. Wolf, *Jumping champions*, Experiment. Math. **8** (1999), no. 2, 107–118, doi:10.1080/10586458.1999.10504393.
- [3] D. A. Goldston and A. H. Ledoan, *Jumping champions and gaps between consecutive primes*, Int. J. Number Theory **7** (2011), no. 6, 1413–1421, doi:10.1142/S179304211100471X.
- [4] D. A. Goldston and A. H. Ledoan, *On the differences between consecutive prime numbers, I*, Integers **12B** (2012/13), Article A3.
- [5] D. A. Goldston and A. H. Ledoan, *On the differences between consecutive prime numbers, II*, preliminary report, AMS Fall Eastern Sectional Meeting, Rochester Institute of Technology, 22–23 September 2012, Abstract 1082-11-179.
- [6] D. A. Goldston and A. H. Ledoan, *The jumping champion conjecture*, Mathematika **61** (2015), no. 3, 719–740, doi:10.1112/S0025579315000078.
- [7] S. Feng and X. Wu, *The k -tuple jumping champions among consecutive primes*, Acta Arith. **156** (2012), no. 4, 325–339, doi:10.4064/aa156-4-2.
- [8] D. A. Goldston and C. Y. Yıldırım, *Higher correlations of divisor sums related to primes III: Small gaps between primes*, Proc. Lond. Math. Soc. (3) **95** (2007), no. 3, 653–686, doi:10.1112/plms/pdm021.
- [9] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [10] F. W. J. Olver, *Asymptotics and Special Functions*, AKP Classics, A K Peters, Wellesley, MA, 1997; reprint of the 1974 Academic Press edition.
- [11] R. Wong, *Asymptotic Approximations of Integrals*, Classics in Applied Mathematics 34, Society for Industrial and Applied Mathematics, Philadelphia, 2001; corrected reprint of the 1989 Academic Press edition.